

The global market share of the two categories is changing inversely. Thirty percent of the two-wheelers sold in China, the world's largest automobile market, are now electric. It is estimated that by 2040, EVs will outsell the ICE cars globally.

Most developed as well as developing countries are now promoting electric mobility, which falls in line with their efforts to reduce emissions and meet their clean energy targets. This could mean the end of combustible engine vehicles by the end of this decade.

Two-wheeler a clear winner

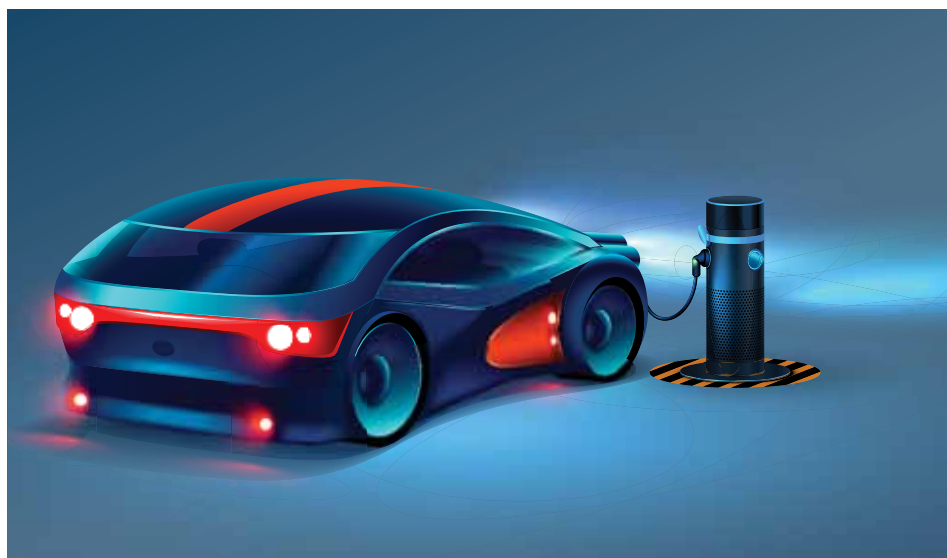
Companies like Tesla have become a shining example of the scope of the electric car industry. But as bankable as it may seem, electric cars need to overcome quite a few hurdles at present to become a profitable business in India.

The Indian consumer is just getting used to the idea of driving a car that runs on batteries, so the shift from an ICE car to an electric car is likely to take some time. As such, the low hanging fruit for India to win big in the EV race is probably going to be the electric two-wheeler and three-wheeler market.

Moreover, the foundation for electric cars is yet to be set in the country. The infrastructure, arguably the most important element, is yet to be developed in India. And given the initial price at which a typical electric car is sold in the market, the average Indian consumer cannot yet be so easily persuaded.

The electric two- and three-wheelers, on the other hand, need much lower infrastructural investment. In terms of their technological advancements and conversion, the market is already witnessing a spike. More importantly, almost 80 percent of the vehicles that run on the Indian roads are two-wheelers or three-wheelers. This makes two- and three-wheelers the perfect segment through which the electrification of vehicles in India can begin.

The prevalence of two-wheelers is not just high in India but all over the world. The Asian market, for instance, is highly reliant on two-wheelers. The boom in the sale of electric models in



countries like China assures that this is a market segment destined to grow.

Projections for electric two-wheelers are estimated to rise consistently, grabbing about 20 percent of the total market by 2026. By 2030, this number is expected to go as high as 60 to 70 percent.

Electric two-wheelers consist of two segments, the low-speed segment (up to 25km/h) and high-speed segment (above 25km/h). The highest inflection point is estimated to be anywhere between the two segments (termed 'city speed' by Hero Electric), while high-speed segment is also expected to rise, albeit as a premium category.

Market pushing for EVs

As the focus is starting to shift towards electric two-wheelers, important aspects are shaping up to be conducive to its growth. The first tailwind that is pushing the growth of the electric two-wheeler market is the efforts of the government to support the industry. Policies like the National Electric Mobility Mission Plan (NEMMP) and Faster Adoption and Manufacturing of Electric Vehicles in India (FAME) Phase 1 and 2 have been launched to push for the change.

Of course, rising climate concerns and the need for clean mobility have been the ultimate harbingers of this mindset. This has led to a growing sense of responsibility for vehicle users who are now exploring the idea of adopting a greener means of transportation like the EVs.

Batteries, for instance, which are undeniably an important element of EVs, have reduced considerably in price. What would cost \$1,200 per kilowatt-hour (kWh) is now available at \$156 dollars per kWh, decreasing by 85 percent. A major reason for this is that these batteries are now being produced for mass usage rather than for mere experimental purposes.

All of this has provided traction to the EV ecosystem. For instance, Hero Electric used to previously struggle to get quality vendors. Now, however, vendors are willing to invest into the category and are looking at it in a serious manner.

Headwinds in EV market

However, the EV ecosystem is still burdened with a few challenges. A considerable speed bump right now is the lack of customer awareness and over-estimation of EVs' performance.

ICE vehicles get an edge with the initial pricing, which is set to be substantially higher for EVs. Even with the same performance, an EV could be 1.8 to 2.8 times more expensive than an ICE vehicle. This difference in performance parity prevents customers from fully understanding the scope of electric two-wheelers, which lies in the total cost of ownership (TCO). The TCO of an electric two-wheeler is much, much lower than that of an ICE one.

Financing is another hurdle. Hero Electric has numerous relationships with non-banking financing com-